

NONSCALAR NON-COMMUTATORS IN ELEMENTARY LINEAR GROUPS

JIALIN LEI AND TENG ZHANG

ABSTRACT. In 2006, Vaserstein asked whether, for some $n > 2$ and some unital ring A , there is an element of $E_n(A)$ which is nonscalar modulo every proper ideal and is not a single commutator. We answer this question affirmatively by establishing two existence theorems. First, for every $n \geq 3$ there is an explicit finite noncommutative algebra A_n over $\mathbb{F}_2$ and an element $g_n \in E_n(A_n)$ with this property; in fact g_n is not a commutator in $\mathrm{GL}_n(A_n)$. The construction is a square-zero extension of a finite product of copies of $\mathbb{F}_2$, and the obstruction is detected after passing along one arrow to the dual numbers over $\mathbb{F}_2$. Second, for every $n \geq 3$ there is also a commutative example, with $A = \mathbb{C}[x]$. The commutative construction uses the unbounded elementary length of $\mathrm{SL}_n(\mathbb{C}[x])$ and the bounded elementary length of linear commutators over commutative rings of finite maximal-spectrum dimension, and then makes the element nonscalar at every closed point by multiplying by a constant elementary transvection.

1. INTRODUCTION

Let A be an associative ring with identity. For $n \geq 2$ and $i \neq j$, we write

$$e_{ij}(a) = I_n + aE_{ij} \quad (a \in A)$$

for the elementary transvection, and denote by $E_n(A)$ the subgroup of $\mathrm{GL}_n(A)$ generated by all such matrices. We use the commutator convention

$$[x, y] = xyx^{-1}y^{-1}.$$

When A is noncommutative, all ideals in this paper are two-sided ideals. A matrix over a quotient A/I is called scalar if it is of the form λI_n for some $\lambda \in A/I$.

The elementary linear group occupies a central place in algebraic K -theory and in the structure theory of linear groups over rings. In the stable range, it is the basic object which measures the difference between matrices and their determinant-like invariants; in the unstable range, it retains delicate information about the ring. One of the most elementary identities in this

2020 *Mathematics Subject Classification*. Primary 20H25; Secondary 20F12, 19B14, 16S70.

Key words and phrases. elementary linear group; commutator; commutator width; square-zero extension; nonscalar modulo ideals; Vaserstein problem.

subject is Steinberg's relation

$$[e_{ik}(a), e_{kj}(1)] = e_{ij}(a), \quad i, j, k \text{ distinct.}$$

Thus, for $n \geq 3$, every elementary generator is itself a commutator. In particular, $E_n(A)$ is generated by commutators in a very strong and explicit sense.

This observation, however, does not decide whether a given element of $E_n(A)$ is a single commutator. The difference between being generated by commutators and being one commutator is substantial. It is the difference between perfectness and commutator width one; it is also the point at which the geometry of the ring begins to intervene. This distinction is already visible in classical work on commutators in linear groups over fields and rings, beginning with Thompson's results for special and general linear groups over fields [12], and continuing through the work of Dennis–Vaserstein [2, 3], Vaserstein–Wheland [13], and Sivatski–Stepanov [10]. In a different direction, bounded elementary generation and commutator width for Chevalley and classical groups have developed into a broad theory; see, for example, [5, 6, 9, 11]. A recurring theme in all of these works is that bounded generation, bounded elementary length, bounded commutator width, and width one are closely related but logically distinct phenomena.

In 2006, Vaserstein posed his *nonscalar commutator problem*; it is also recorded in the Kourovka Notebook [8, p. 97, Problem 16.23].

Problem 1.1 (Vaserstein). *Is there, for some $n > 2$ and some unital ring A , an element of $E_n(A)$ which is nonscalar modulo every proper ideal of A and is not a commutator?*

The condition that the element remain nonscalar modulo every proper ideal is the essential feature of Problem 1.1. Without it, local square-zero constructions already give non-commutator phenomena in special linear groups; compare Avni–Gelander–Kassabov–Shalev [1, Proposition 3.2, Corollary 3.3, Theorem 3.5]. Such examples show that width one can fail, but they do not answer Problem 1.1, because the obstruction may disappear in quotients. Vaserstein's hypothesis rules out this degeneracy. It asks for an element which is nonscalar in every quotient of the ring and yet still cannot be written as one commutator. Thus the question isolates a genuinely global obstruction to width one.

For commutative rings, the condition has a geometric interpretation. If $A = \mathbb{C}[x]$, then a matrix is nonscalar modulo every proper ideal precisely when it is nonscalar at every closed point of the affine line. Thus Problem 1.1 asks for a polynomial family of elementary matrices which avoids the scalar locus at every point, but which nevertheless lies outside the image of the commutator map

$$\mathrm{GL}_n(A) \times \mathrm{GL}_n(A) \longrightarrow \mathrm{GL}_n(A), \quad (x, y) \longmapsto [x, y].$$

This formulation places the problem at the boundary between linear algebra over rings, unstable K -theory, and the geometry of representation varieties.

The purpose of this paper is to answer Problem 1.1 affirmatively. In fact, we prove two results, one finite and noncommutative, the other commutative.

Theorem 1.2. *For every integer $n \geq 3$, there is a finite noncommutative unital ring A_n and an element $g_n \in E_n(A_n)$ such that g_n is nonscalar modulo every proper two-sided ideal of A_n , and g_n is not a commutator in $\mathrm{GL}_n(A_n)$.*

The conclusion of Theorem 1.2 is stronger than the formulation of Problem 1.1: the element is not a commutator even in the ambient group $\mathrm{GL}_n(A_n)$, not merely in $E_n(A_n)$. The ring A_n is an explicit finite algebra over $\mathbb{F}_2$. It is a square-zero extension of a finite product of copies of $\mathbb{F}_2$, built from the finite set of all commutator presentations of a fixed transvection in $\mathrm{GL}_n(\mathbb{F}_2)$. The obstruction is detected by isolating one arrow and passing to the dual numbers over $\mathbb{F}_2$.

Our second result shows that the phenomenon is not an artifact of non-commutativity.

Theorem 1.3. *For every integer $n \geq 3$, there is an element $g_n \in E_n(\mathbb{C}[x])$ such that g_n is nonscalar modulo every proper ideal of $\mathbb{C}[x]$, and g_n is not a commutator in $\mathrm{GL}_n(\mathbb{C}[x])$.*

This gives a commutative answer over one of the most classical rings. The proof is necessarily of a different nature. Since $\mathbb{C}[x]$ is Euclidean,

$$E_n(\mathbb{C}[x]) = \mathrm{SL}_n(\mathbb{C}[x]).$$

The construction uses two deep facts about elementary length. Van der Kallen [7] proved that $\mathrm{SL}_3(\mathbb{C}[x])$ is not boundedly generated by elementary matrices, and Erovenko [4] gave an explicit proof which, in particular, applies to $\mathrm{SL}_n(F[x])$ for $n \geq 3$ when F has infinite transcendence degree over its prime subfield. On the other hand, Sivatski and Stepanov [10] proved that linear commutators over commutative rings of finite maximal-spectrum dimension have bounded elementary length. Combining these two facts gives an element $h \in E_n(\mathbb{C}[x])$ which is not a product of two $\mathrm{GL}_n(\mathbb{C}[x])$ -commutators. A short specialization-avoidance argument then modifies h by a constant elementary transvection so that the resulting element is nonscalar at every closed point, while preserving the obstruction to being a single commutator.

Organization of the paper. In Section 2 we record the elementary square-zero lemma used in the finite construction. In particular, we isolate a dual-number specialization associated to one arrow of a square-zero extension of a finite product of copies of $\mathbb{F}_2$. Section 3 proves Theorem 1.2. We construct the finite algebra A_n , prove that the chosen element lies in $E_n(A_n)$, verify the nonscalar condition in all quotients, and establish the determinant obstruction. Section 4 proves Theorem 1.3. We first use unbounded elementary length and bounded elementary length of commutators to produce a long element of $E_n(\mathbb{C}[x])$, and then apply the finite specialization-avoidance argument.

Acknowledgements and AI tools disclosure. The authors are deeply grateful to members of the community working on this topic for their helpful comments. This paper was completed in April 2026, and no AI tools were used in its preparation.

2. PRELIMINARIES

We record two elementary facts used repeatedly.

Lemma 2.1. *Let F be a field, let Ω be a finite set, and let A be the F -algebra with basis*

$$\{p_\omega : \omega \in \Omega\} \cup \{a_{\omega\eta} : \omega, \eta \in \Omega, \omega \neq \eta\}$$

and multiplication

$$p_\omega p_\eta = \delta_{\omega\eta} p_\omega,$$

$$p_\zeta a_{\omega\eta} = \delta_{\zeta\omega} a_{\omega\eta}, \quad a_{\omega\eta} p_\zeta = \delta_{\eta\zeta} a_{\omega\eta},$$

and

$$a_{\omega\eta} a_{\zeta\theta} = 0.$$

Then A is an associative unital algebra with unit $1_A = \sum_{\omega \in \Omega} p_\omega$. If

$$J = \text{span}_F \{a_{\omega\eta} : \omega \neq \eta\},$$

then $J^2 = 0$ and $A/J \cong \prod_{\omega \in \Omega} F$. Moreover, if $u, v \in M_n(A)$ are written uniquely as

$$u = \sum_{\nu \in \Omega} p_\nu u_\nu + \sum_{\alpha \neq \beta} a_{\alpha\beta} u_{\alpha\beta}, \quad v = \sum_{\nu \in \Omega} p_\nu v_\nu + \sum_{\alpha \neq \beta} a_{\alpha\beta} v_{\alpha\beta},$$

with coefficients in $M_n(F)$, then

$$(2.1) \quad (uv)_{\omega\eta} = u_\omega v_{\omega\eta} + u_{\omega\eta} v_\eta.$$

Proof. Associativity is verified on basis elements. Products containing two arrows are zero. Products containing no arrow are governed by orthogonal idempotents. Products containing exactly one arrow reduce on both sides to the same source and target Kronecker symbols. The stated unit and the identities $J^2 = 0$ and $A/J \cong \prod_{\Omega} F$ are immediate. Formula (2.1) is the matrix form of the same source-target multiplication rule. $\square$

Lemma 2.2. *In the situation of Lemma 2.1, fix distinct vertices $\omega, \eta \in \Omega$ and define*

$$S_{\omega\eta} = \{z \in M_n(A) : z_\omega = z_\eta\}.$$

Then $S_{\omega\eta}$ is a unital subring of $M_n(A)$, and

$$\Theta_{\omega\eta} : S_{\omega\eta} \longrightarrow M_n(F[\varepsilon]/(\varepsilon^2)), \quad \Theta_{\omega\eta}(z) = z_\omega + \varepsilon z_{\omega\eta},$$

is a unital ring homomorphism. If $z \in S_{\omega\eta}$ is invertible in $M_n(A)$, then $z^{-1} \in S_{\omega\eta}$.

Proof. The set $S_{\omega\eta}$ is plainly closed under addition and additive inverses, and it contains the identity matrix. If $u, v \in S_{\omega\eta}$, then

$$(uv)_{\omega} = u_{\omega}v_{\omega}, \quad (uv)_{\eta} = u_{\eta}v_{\eta} = u_{\omega}v_{\omega},$$

so $S_{\omega\eta}$ is closed under multiplication. Formula (2.1), together with $v_{\eta} = v_{\omega}$, gives

$$\Theta_{\omega\eta}(uv) = u_{\omega}v_{\omega} + \varepsilon(u_{\omega}v_{\omega\eta} + u_{\omega\eta}v_{\omega}) = \Theta_{\omega\eta}(u)\Theta_{\omega\eta}(v).$$

Thus $\Theta_{\omega\eta}$ is a unital ring homomorphism. Finally, if z is invertible and $z_{\omega} = z_{\eta}$, then

$$(z^{-1})_{\omega} = (z_{\omega})^{-1} = (z_{\eta})^{-1} = (z^{-1})_{\eta},$$

so $z^{-1} \in S_{\omega\eta}$. □

3. THE FINITE NONCOMMUTATIVE CONSTRUCTION

Fix $n \geq 3$ and put

$$F = \mathbb{F}_2, \quad H = \mathrm{GL}_n(F), \quad t = e_{12}(1) = I_n + E_{12} \in H.$$

Let

$$D = \{(h, \ell) \in H^2 : [h, \ell] = t\}.$$

The set D is finite and nonempty, because

$$[e_{13}(1), e_{32}(1)] = e_{12}(1) = t.$$

Set $\Omega = D \sqcup \{\infty\}$. Thus $|\Omega| = |D| + 1$. Let $A = A_n$ be the F -algebra attached to Ω by Lemma 2.1. Since Ω is finite, A is a finite-dimensional F -algebra; in particular, A is a finite ring. Moreover, A is noncommutative. Indeed, D is nonempty, so $|\Omega| \geq 2$. Choose two distinct vertices $\omega, \eta \in \Omega$. Then

$$p_{\omega}a_{\omega\eta} = a_{\omega\eta}, \quad a_{\omega\eta}p_{\omega} = 0.$$

Hence $p_{\omega}a_{\omega\eta} \neq a_{\omega\eta}p_{\omega}$, and so A is noncommutative.

Put

$$r = \sum_{\omega \neq \eta} a_{\omega\eta} \in J.$$

Since $J^2 = 0$, one has $r^2 = 0$. Define

$$(3.1) \quad g = (I_n + rE_{11})e_{12}(1_A) \in \mathrm{GL}_n(A).$$

Proposition 3.1. *The element g defined in (3.1) belongs to $E_n(A)$.*

Proof. It is enough to prove that $I_n + rE_{11} \in E_n(A)$, since $e_{12}(1_A)$ is elementary. Fix an arrow $a = a_{\omega\eta}$ and put $p = p_{\eta}$. Then

$$ap = a, \quad pa = 0, \quad a^2 = 0.$$

Let $X = aE_{12}$ and $Y = pE_{21}$. Since $X^2 = Y^2 = 0$ and the characteristic is 2, the inverses of $I_n + X$ and $I_n + Y$ are themselves. Moreover

$$XY = aE_{11}, \quad YX = 0,$$

and all higher alternating products XYX , YXY , and $(XY)^2$ vanish. Hence

$$[I_n + X, I_n + Y] = (I_n + X)(I_n + Y)(I_n + X)(I_n + Y) = I_n + aE_{11}.$$

Both $I_n + X$ and $I_n + Y$ are elementary matrices; therefore $I_n + aE_{11} \in E_n(A)$. Multiplying this identity over all arrows, in any order, gives

$$\prod_{\omega \neq \eta} (I_n + a_{\omega\eta} E_{11}) = I_n + rE_{11},$$

because all products of two arrows are zero. Thus $I_n + rE_{11} \in E_n(A)$. $\square$

Proposition 3.2. *For every proper two-sided ideal $I \triangleleft A$, the image of g in $M_n(A/I)$ is not scalar.*

Proof. Expanding (3.1),

$$g = I_n + E_{12} + rE_{11} + rE_{12}.$$

Thus the $(1, 2)$ -entry of g is $1_A + r$. Since $r^2 = 0$ and $\text{char } A = 2$,

$$(1_A + r)^2 = 1_A.$$

Hence $1_A + r$ is a unit. No proper ideal contains a unit, so $1_A + r \notin I$ for every proper ideal I . The image of the $(1, 2)$ -entry of g in A/I is therefore nonzero. A scalar matrix has zero off-diagonal entries, and the claim follows. $\square$

Proposition 3.3. *The element g is not a commutator in $\text{GL}_n(A)$.*

Proof. Assume, toward a contradiction, that

$$g = [x, y]$$

for some $x, y \in \text{GL}_n(A)$. Reducing modulo J , and using $A/J \cong \prod_{\omega \in \Omega} F$, gives matrices

$$x_\omega, y_\omega \in \text{GL}_n(F) \quad (\omega \in \Omega).$$

Since r maps to zero in A/J , the reduction of g at every vertex is $t = e_{12}(1)$. Hence

$$[x_\omega, y_\omega] = t$$

for all $\omega \in \Omega$, so $(x_\omega, y_\omega) \in D$ for all ω . Since $|\Omega| = |D| + 1$, there are two distinct vertices $\omega \neq \eta$ such that

$$(x_\omega, y_\omega) = (x_\eta, y_\eta).$$

In particular $x, y \in S_{\omega\eta}$, and by Lemma 2.2 also $x^{-1}, y^{-1} \in S_{\omega\eta}$. Applying $\Theta = \Theta_{\omega\eta}$ gives

$$\Theta(g) = [\Theta(x), \Theta(y)]$$

in $\text{GL}_n(F[\varepsilon]/(\varepsilon^2))$. Since $F[\varepsilon]/(\varepsilon^2)$ is commutative, determinants of group commutators are equal to 1. Thus

$$(3.2) \quad \det \Theta(g) = 1.$$

We now compute $\Theta(g)$. Put

$$s = I_n + rE_{11}, \quad e = e_{12}(1_A).$$

Then $g = se$. At every vertex ν one has

$$s_\nu = I_n, \quad e_\nu = t,$$

so $g_\nu = t$. Along the chosen arrow $a_{\omega\eta}$ one has

$$s_{\omega\eta} = E_{11}, \quad e_{\omega\eta} = 0.$$

Using (2.1),

$$g_{\omega\eta} = s_\omega e_{\omega\eta} + s_{\omega\eta} e_\eta = I_n \cdot 0 + E_{11}t = E_{11}t.$$

Therefore

$$\Theta(g) = t + \varepsilon E_{11}t = (I_n + \varepsilon E_{11})t.$$

Since $t = e_{12}(1)$ is elementary, $\det(t) = 1$, and hence

$$\det \Theta(g) = \det(I_n + \varepsilon E_{11}) \det(t) = 1 + \varepsilon.$$

But $1 + \varepsilon \neq 1$ in $F[\varepsilon]/(\varepsilon^2)$, contradicting (3.2). Thus g is not a commutator in $\mathrm{GL}_n(A)$. $\square$

Proof of Theorem 1.2. Combine Propositions 3.1, 3.2, and 3.3. $\square$

4. THE COMMUTATIVE CONSTRUCTION OVER $\mathbb{C}[x]$

We now turn to commutative rings. In this section $A = \mathbb{C}[x]$ and $n \geq 3$ is fixed. Since $\mathbb{C}[x]$ is Euclidean,

$$E_n(\mathbb{C}[x]) = \mathrm{SL}_n(\mathbb{C}[x]).$$

The proof is the usual elementary row reduction over a Euclidean domain.

4.1. A long element. For a group G and a subset $X \subseteq G$, write $\ell_X(g)$ for the word length of g with respect to $X \cup X^{-1}$. Let

$$\mathcal{E}_n(R) = \{e_{ij}(a) : i \neq j, a \in R\}$$

be the set of elementary transvections in $E_n(R)$.

We shall use the following two length lemmas. The first is an explicit form of van der Kallen's theorem, due to Erovenko [4, p. 163, Theorem 2.1]. The hypothesis of that theorem is satisfied because $\mathbb{C}$ has infinite transcendence degree over its prime subfield.

Lemma 4.1 ([4, Theorem 2.1]). *The group $\mathrm{SL}_n(\mathbb{C}[x])$ has unbounded word length with respect to the set of elementary transvections. Equivalently, for every integer $N \geq 1$ there exists an element $h \in \mathrm{SL}_n(\mathbb{C}[x])$ which is not a product of at most N elementary transvections.*

The second lemma is the bounded-length theorem of Sivatski and Stepanov. We use it only for the ring $\mathbb{C}[x]$. In their notation, the relevant general result is [10, p. 296, Theorem 1.1]: if R is a Noetherian commutative ring with $\dim \mathrm{Max} R = d < \infty$, then for every $n \geq 3$ there is a constant depending only on n and d which bounds the elementary length of all commutators

$$[a, b], \quad a \in E_n(R), \quad b \in \mathrm{SL}_n(R).$$

Since $\mathbb{C}[x]$ is Noetherian and $\dim \mathrm{Max} \mathbb{C}[x] = 1$, this gives the following form.

Lemma 4.2 ([10, Theorem 1.1]). *There exists a constant $L = L(n)$ such that every commutator*

$$[a, b], \quad a \in E_n(\mathbb{C}[x]), \quad b \in \mathrm{SL}_n(\mathbb{C}[x]),$$

is a product of at most L elementary transvections.

Proposition 4.3. *There exists $h \in E_n(\mathbb{C}[x])$ which is not a product of two commutators in $\mathrm{GL}_n(\mathbb{C}[x])$.*

Proof. Let $L = L(n)$ be the constant from Lemma 4.2. We first show that every commutator in $\mathrm{GL}_n(\mathbb{C}[x])$ has elementary length at most L .

Let $[u, v]$ be an arbitrary commutator in $\mathrm{GL}_n(\mathbb{C}[x])$. Since the only units of $\mathbb{C}[x]$ are the nonzero constants, both $\det(u)$ and $\det(v)$ belong to $\mathbb{C}^*$. Choose $\alpha, \beta \in \mathbb{C}^*$ such that

$$\alpha^n \det(u) = 1, \quad \beta^n \det(v) = 1.$$

Put $u' = (\alpha I_n)u$ and $v' = (\beta I_n)v$. Then $u', v' \in \mathrm{SL}_n(\mathbb{C}[x])$. Since $\mathbb{C}[x]$ is Euclidean, we have

$$\mathrm{SL}_n(\mathbb{C}[x]) = E_n(\mathbb{C}[x]).$$

Thus $u' \in E_n(\mathbb{C}[x])$ and $v' \in \mathrm{SL}_n(\mathbb{C}[x])$. Scalar matrices are central, so

$$[u, v] = [u', v'].$$

By Lemma 4.2, the commutator $[u', v']$ is a product of at most L elementary transvections. Hence every commutator in $\mathrm{GL}_n(\mathbb{C}[x])$ has elementary length at most L .

Now suppose, for contradiction, that every element of $E_n(\mathbb{C}[x])$ were a product of two commutators in $\mathrm{GL}_n(\mathbb{C}[x])$. Then every element of $E_n(\mathbb{C}[x])$ would be a product of at most $2L$ elementary transvections. Since

$$E_n(\mathbb{C}[x]) = \mathrm{SL}_n(\mathbb{C}[x]),$$

this contradicts Lemma 4.1. Therefore there exists $h \in E_n(\mathbb{C}[x])$ which is not a product of two commutators in $\mathrm{GL}_n(\mathbb{C}[x])$. $\square$

4.2. Avoiding scalar specializations. For $\lambda \in \mathbb{C}$ with $\lambda^n = 1$, let

$$L_\lambda = \{\lambda(I_n - dE_{12}) : d \in \mathbb{C}\} \subseteq \mathrm{SL}_n(\mathbb{C}).$$

Set

$$U_n = \bigcup_{\lambda^n=1} L_\lambda.$$

This is a finite union of affine lines in $M_n(\mathbb{C})$.

Lemma 4.4. *Let $h \in E_n(\mathbb{C}[x])$ be as in Proposition 4.3. Then the polynomial map*

$$h : \mathbb{A}_{\mathbb{C}}^1 \longrightarrow \mathrm{SL}_n(\mathbb{C})$$

has image not contained in U_n .

Proof. Suppose that $h(\mathbb{A}_{\mathbb{C}}^1) \subseteq U_n$. Since $\mathbb{A}_{\mathbb{C}}^1$ is irreducible and U_n is a finite union of closed affine lines, the image is contained in one of them. Thus there are $\lambda \in \mathbb{C}^*$ with $\lambda^n = 1$ and $f \in \mathbb{C}[x]$ such that

$$h = \lambda(I_n - fE_{12}).$$

We show that the right-hand side is a product of two $\mathrm{GL}_n(\mathbb{C}[x])$ -commutators.

First,

$$I_n - fE_{12} = e_{12}(-f) = [e_{13}(-f), e_{32}(1)].$$

Second, λI_n is a commutator in $\mathrm{GL}_n(\mathbb{C})$. Indeed, let

$$D = \mathrm{diag}(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$$

and let P be the permutation matrix defined by $Pe_i = e_{i+1}$ for $1 \leq i < n$ and $Pe_n = e_1$. A direct calculation gives

$$DPD^{-1}P^{-1}e_i = \lambda e_i \quad (1 \leq i \leq n),$$

where for $i = 1$ one uses $\lambda^n = 1$. Hence

$$[D, P] = \lambda I_n.$$

Therefore h is a product of two commutators in $\mathrm{GL}_n(\mathbb{C}[x])$, contradicting Proposition 4.3. Hence $h(\mathbb{A}_{\mathbb{C}}^1)$ is not contained in U_n . $\square$

Lemma 4.5. *There exists $c \in \mathbb{C}$ such that $g = e_{12}(c)h$ is nonscalar at every closed point of $\mathbb{A}_{\mathbb{C}}^1$; equivalently, $g(a)$ is not a scalar matrix for every $a \in \mathbb{C}$.*

Proof. By Lemma 4.4, the closed set $h^{-1}(U_n) \subseteq \mathbb{A}_{\mathbb{C}}^1$ is proper. Hence it is finite. If

$$e_{12}(c)h(a) = \mu I_n$$

for some $a \in \mathbb{C}$, then

$$h(a) = e_{12}(-c)\mu I_n = \mu(I_n - cE_{12}).$$

Because $h(a) \in \mathrm{SL}_n(\mathbb{C})$, one has $\mu^n = 1$, so $h(a) \in U_n$. Therefore all possible bad pairs (a, c) have $a \in h^{-1}(U_n)$. For a fixed such a , there is at most one corresponding value of c : indeed, if $h(a) = \mu(I_n - cE_{12})$, then $\mu = h(a)_{11}$ and $c = -\mu^{-1}h(a)_{12}$. Hence there are only finitely many bad values of c . Choose $c \in \mathbb{C}$ outside this finite set. $\square$

Proposition 4.6. *For c chosen as in Lemma 4.5, the element $g = e_{12}(c)h$ belongs to $E_n(\mathbb{C}[x])$, is nonscalar modulo every proper ideal of $\mathbb{C}[x]$, and is not a commutator in $\mathrm{GL}_n(\mathbb{C}[x])$.*

Proof. Since both $e_{12}(c)$ and h belong to $E_n(\mathbb{C}[x])$, so does g .

Let $I \triangleleft \mathbb{C}[x]$ be a proper ideal. If the image of g in $M_n(\mathbb{C}[x]/I)$ were scalar, then, choosing a maximal ideal $(x - a)$ containing I , its further image in $M_n(\mathbb{C})$ would also be scalar. That image is $g(a)$, contradicting Lemma 4.5. Hence g is nonscalar modulo every proper ideal.

Finally suppose that $g = [u, v]$ for some $u, v \in \mathrm{GL}_n(\mathbb{C}[x])$. Since

$$e_{12}(-c) = [e_{13}(-c), e_{32}(1)],$$

we get

$$h = e_{12}(-c)g$$

as a product of two $\mathrm{GL}_n(\mathbb{C}[x])$ -commutators. This contradicts the choice of h in Proposition 4.3. Therefore g is not a commutator in $\mathrm{GL}_n(\mathbb{C}[x])$. $\square$

Proof of Theorem 1.3. This is Proposition 4.6. $\square$

Finally, let us give some concluding remarks. The two constructions in this paper are formally different. The noncommutative example is finite and explicit. Its obstruction is a determinant obstruction only after a carefully chosen square-zero arrow is isolated; the ambient ring itself is noncommutative, but the selected arrow quotient lands in the commutative dual numbers. The commutative example has the opposite character: the ring $\mathbb{C}[x]$ is classical, but the existence of the initial long element uses the deep fact that elementary length in $\mathrm{SL}_n(\mathbb{C}[x])$ is unbounded whereas commutators have bounded elementary length.

In both cases the conclusion is stronger than the literal negative assertion about $E_n(A)$: the constructed element is not a commutator even in the larger group $\mathrm{GL}_n(A)$. Moreover, both constructions work uniformly for every fixed $n \geq 3$.

REFERENCES

- [1] N. Avni, T. Gelander, M. Kassabov, and A. Shalev, *Word values in p -adic and adelic groups*, Bull. Lond. Math. Soc. **45** (2013), no. 6, 1323–1330. doi:[10.1112/blms/bdt063](https://doi.org/10.1112/blms/bdt063)
- [2] R. K. Dennis and L. N. Vaserstein, *On a question of M. Newman on the number of commutators*, J. Algebra **118** (1988), no. 1, 150–161. doi:[10.1016/0021-8693\(88\)90055-5](https://doi.org/10.1016/0021-8693(88)90055-5)
- [3] R. K. Dennis and L. N. Vaserstein, *Commutators in linear groups*, K-Theory **2** (1989), no. 6, 761–767. doi:[10.1007/BF00538432](https://doi.org/10.1007/BF00538432)
- [4] I. V. Erovenko, *$\mathrm{SL}_n(F[x])$ is not boundedly generated by elementary matrices: explicit proof*, Electron. J. Linear Algebra **11** (2004), 162–167. doi:[10.13001/1081-3810.1129](https://doi.org/10.13001/1081-3810.1129)
- [5] P. Gvozdevsky, *Verbal width in arithmetic Chevalley groups*, J. Algebra **666** (2025), 82–93. doi:[10.1016/j.jalgebra.2024.12.002](https://doi.org/10.1016/j.jalgebra.2024.12.002)
- [6] R. Hazrat, A. Stepanov, N. Vavilov, and Z. Zhang, *Commutator width in Chevalley groups*, Note Mat. **33** (2013), no. 1, 139–170. doi:[10.1285/i15900932v33n1p139](https://doi.org/10.1285/i15900932v33n1p139)
- [7] W. van der Kallen, *$\mathrm{SL}_3(\mathbb{C}[X])$ does not have bounded word length*, in *Algebraic K-theory, Part I (Oberwolfach, 1980)*, Lecture Notes in Math., vol. 966, Springer, Berlin, 1982, pp. 357–361. doi:[10.1007/BFb0062183](https://doi.org/10.1007/BFb0062183)
- [8] E. I. Khukhro and V. D. Mazurov (eds.), *The Kourovka Notebook: Unsolved Problems in Group Theory*, 21st ed., arXiv:1401.0300v44 [math.GR], 2026. doi:[10.48550/arXiv.1401.0300](https://doi.org/10.48550/arXiv.1401.0300)
- [9] B. Kunyavskii, E. Plotkin, and N. Vavilov, *Bounded generation and commutator width of Chevalley groups: function case*, Eur. J. Math. **9** (2023), no. 3, Art. No. 53. doi:[10.1007/s40879-023-00627-y](https://doi.org/10.1007/s40879-023-00627-y)
- [10] A. S. Sivatski and A. V. Stepanov, *On the word length of commutators in $\mathrm{GL}_n(R)$* , K-Theory **17** (1999), no. 4, 295–302. doi:[10.1023/A:1007730801851](https://doi.org/10.1023/A:1007730801851)
- [11] A. Stepanov and N. Vavilov, *On the length of commutators in Chevalley groups*, Israel J. Math. **185** (2011), 253–276. doi:[10.1007/s11856-011-0109-2](https://doi.org/10.1007/s11856-011-0109-2)
- [12] R. C. Thompson, *Commutators in the special and general linear groups*, Trans. Amer. Math. Soc. **101** (1961), 16–33. doi:[10.1090/S0002-9947-1961-0130917-7](https://doi.org/10.1090/S0002-9947-1961-0130917-7)

- [13] L. N. Vaserstein and E. Wheland, *Commutators and companion matrices over rings of stable rank 1*, Linear Algebra Appl. **142** (1990), 263–277. doi:[10.1016/0024-3795\(90\)90270-M](https://doi.org/10.1016/0024-3795(90)90270-M)

DEPARTMENT OF MATHEMATICS, BINGHAMTON UNIVERSITY, BINGHAMTON, NY 13902,
USA

Email address: `jlei15@binghamton.edu`

SCHOOL OF MATHEMATICS AND STATISTICS, XI'AN JIAOTONG UNIVERSITY, XI'AN
710049, P. R. CHINA

Email address: `teng.zhang@stu.xjtu.edu.cn`